Improving Delivery Efficiency Through Driver-Led Stop Reordering at DHL

Roman Finger, Serguei Anghelov, Murat Tutar

Updated on July 18, 2025

1 Introduction

In modern urban parcel delivery, unpredictable traffic often causes delays, higher fuel use, and lower customer satisfaction. Traditional static route-planning cannot adapt to sudden congestion. To address this challenge, we propose an eight-week field experiment to give drivers limited autonomy to reorder upcoming stops based on live traffic data. Our core research question is:

> Does allowing drivers to dynamically reorder their next three stops – thus deviating from the pre-calculated sequence of stops, which is expected to be optimal under normal, static conditions – improve delivery efficiency without compromising customer satisfaction?

The study will be conducted within DHL's non-perishable e-commerce delivery operations, where minor stop reordering does not affect product quality. We assess whether dynamic adjustments improve on-time delivery or reduce total route duration, while maintaining customer satisfaction (measured by punctuality and feedback).

From a business perspective, small time savings can yield meaningful gains. Saving 3–5 minutes per route could enable extra deliveries per driver each week, reducing overtime and fuel use. For example, if 100 drivers each save 4 minutes per shift, that's roughly 400 minutes (6.7 hours) saved daily. At €20 per driver-hour, this equals over €1300 saved per day accumulating to substantial annual savings. We will calculate a simple ROI by comparing projected annual savings to the one-time costs of enabling reordering and briefing drivers. This will help assess whether efficiency gains justify implementation costs.

While dynamic routing is well-established in theory, field evidence remains limited. Similarly, empowering drivers fits with participative decision-making theory—enhancing problem-solving under uncertainty—but excessive flexibility risks errors. Our design balances autonomy and structure (e.g., limiting

reordering to the next three stops) and provides an intuitive interface. We will monitor key metrics (on-time delivery, first-attempt success, customer satisfaction ratings) to ensure efficiency gains do not come at the cost of service quality.

2 Experimental Design

2.1 Model and Hypotheses

The treatment, which is allowing drivers to reorder their next three delivery stops, directly influences driver routing decisions. These decisions, in turn, impact operational outcomes such as total route duration and number of stops completed, as well as service outcomes like on-time delivery rate and customer satisfaction.

We formalize the intervention's logic as follows: Let $T_i = 1$ if driver i is assigned to the treatment group (permitted to reorder stops) and $T_i = 0$ if assigned to the control group (required to follow the original sequence). Let Y_i be the total route duration for driver i in a shift and let S_i be a composite service score (e.g., on-time delivery percentage or average satisfaction rating). We also collect controls X_i (route length, driver experience, vehicle type, shift start time, route environment (urban/rural), day of week, weather condition). We specify two regression models:

$$Y_i = \alpha + \beta T_i + \delta X_i + \epsilon_i$$
$$S_i = \gamma + \theta T_i + \lambda X_i + \mu_i$$

Here, β (expected negative) measures the treatment effect on efficiency (lower route time means improved efficiency), and θ (expected near zero or positive) measures the effect on service quality.

We derive two hypotheses:

- **H1 (Efficiency):** Treatment drivers (with reordering autonomy) will achieve higher operational efficiency, i.e., lower route duration ($\beta < 0$). Rationale: Rerouting to avoid congestion should reduce idle time and speed up delivery.
- **H2 (Service):** The treatment will not worsen customer satisfaction ($\theta \approx 0$). Rationale: Delays at some stops are offset by time savings elsewhere, keeping service outcomes stable.

Key assumptions include: drivers comply with their assigned condition (treatment can reorder; control cannot), random assignment ensures T_i is independent of unobservables, and all drivers are competent in using navigation, so route comprehension does not limit outcomes during the eight-week trial.

2.2 Sample Selection and Randomization

We will recruit 100–120 full-time delivery drivers from one urban depot of the parcel company. This sample (∼50–60 per group) is based on a priori power calculations targeting 80% power to detect a moderate effect (Cohen's $d \approx 0.5$) at a 5% significance level, balancing sensitivity and feasibility. Over eight weeks, each driver will complete roughly 40 shifts across varied traffic and weather conditions (peak/off-peak, weekdays/weekends).

Drivers will be randomly assigned to treatment or control at the trial's start (no crossover). To ensure balance across key variables influencing route complexity, we apply stratified randomization based on four binary factors: (1) route area (urban vs. suburban), (2) shift start time (before vs. after 9:00 AM), (3) driver tenure (¡1 year vs. ≥1 year), and (4) stop density (¡5 vs. ≥5 stops/km). These categories are supported by company data as drivers of route time and workload.

2.3 Treatment and Control Groups

The company already provides drivers with a mobile routing app for planning, i.e. every driver has an algorithmic based route suggestion given by an already-on-use app used by company. In the control group, drivers follow the standard procedure: they receive the planned route sequence (generated by the existing system) and are instructed to follow it exactly. No reordering tools are available to them; any deviation is allowed only for true emergencies (e.g., road closures).

In the treatment group, drivers receive the same pre-determined optimal route as the control group, based on historical data and standard planning procedures. However, they are permitted to deviate from this sequence by reordering their next three stops during each shift if they believe this will improve efficiency, such as by avoiding traffic congestion. All other conditions, including total load, delivery time windows, and break schedules, remain the same between the two groups. Drivers receive instructions regarding when and how to exercise this discretion prior to the trial.

2.4 Procedure, Data Collection, and Ethics

Drivers check in and out via the app, which logs shift times. GPS data records route details, including stop order and timestamps. A manipulation check will compare actual vs. planned sequences, with more reordering expected in the treatment group. We will collect data at the driver-shift level:

- Operational data: Stop arrival/departure times, total route distance, number of stops completed.

- Service data: On-time delivery (within promised window) and success on first attempt.
- Customer feedback: Standard survey rating (1–10) and any logged complaint.
- Control variables: total parcel volume per route, vehicle type, day of week, and weather.

These will be used in analysis to adjust for remaining differences.

All participants will give informed consent. In line with German Data Protection Regulations (GDPR), drivers are informed about GPS tracking and data use; all personal identifiers are anonymized. Only job-related data are securely stored, and only aggregated metrics are reported to ensure confidentiality. Drivers are told the experiment is for learning and improvement, not performance evaluation, and participation is voluntary.

3 Data Analysis

This section describes how we will transform raw logs into analyzable data, test our hypotheses, and translate results into business-relevant terms. Analyses are conducted at the driver-shift level (one row per driver per completed shift), with supplementary delivery-level diagnostics where needed.

3.1 Data Structure & Preparation

For each driver-shift we will assemble:

- **Route Duration (minutes):** checkout time minus checkin time, adjusted for scheduled break periods if logged separately.
- **Stops Completed:** count of parcels successfully delivered.
- **Distance Driven (km):** from GPS trace; used as a control and for productivity ratios (stops/km, minutes/km).
- **On-Time Deliveries (%):** share of stops delivered within promised time windows.
- **First-Attempt Success (%):** share of parcels delivered without reattempt.
- **Customer Rating (mean 1–10):** average of survey responses received within 24h of delivery (if $\geq$1 response); missing otherwise.
- **Complaint Flag:** 1 if any formal complaint logged for that shift's deliveries; else 0.

- **Reorder Count:** number of deviations from the planned sequence; used for manipulation checks and exploratory mediation.
- **Control variables:** route environment (urban/suburban), shift start (early/late), driver tenure (¡ 1 year vs $\geq$ 1 year), stop density (¡5 stops/km vs $\geq$5 stops/km), parcel volume, vehicle type, day of week, and severe weather indicator.

3.2 Outcome Hierarchy: Primary vs. Secondary Metrics

To focus inference and manage multiple comparisons, we prespecify:

- **Primary efficiency outcome:** Route Duration (minutes).
- **Primary service outcome:** On-Time Deliveries (%).
- **Key secondary outcomes:** First-Attempt Success (%), Customer Rating, Stops Completed, Minutes per Stop, and Complaint Flag.

All reported p-values for primary outcomes use $\alpha = 0.05$ (two-tailed). Secondary outcomes are interpreted descriptively and with false-discovery caution.

3.3 Descriptive Diagnostics

We will produce baseline tables comparing treatment and control drivers (pre-trial characteristics) to confirm successful randomization. For study data, we will summarize each outcome by group: means, standard deviations, medians, and interquartile ranges (continuous); proportions and counts (binary). Histograms and boxplots will display distributional differences; weekly trend plots will check for time drift or learning.

3.4 Hypothesis Tests

H1 Efficiency (Route Duration):

First, we will assess whether the distribution of route durations approximates normality using the Shapiro–Wilk test (with $p > 0.05$ indicating normality) and visual tools such as Q–Q plots.

Then, if normality holds, we will apply a two-sample t-test to compare the mean route durations between treatment and control groups. If the normality assumption is violated, we will instead use the Mann–Whitney U test to compare the median route durations.

In either case, we will report the estimated mean (or median) difference in minutes, the corresponding 95% confidence interval, the test statistic, and the p-value.

H2 Service (On-Time Percentage):

We will compare the mean on-time delivery rates between the two groups.

If the on-time rates are approximately normally distributed, we will use a two-sample t-test. Otherwise, we will employ a nonparametric alternative (Mann–Whitney U) to compare central tendencies.

To aid interpretation for this binary-outcome metric, we will also perform a two-proportion z-test to estimate the absolute difference in on-time rates (in percentage points) along with a 95% confidence interval.

Secondary Metrics:

For other continuous outcomes (e.g., average customer rating, minutes per stop), we will follow the same normality-check–then–t-test or Mann–Whitney U procedure. For other binary outcomes (e.g., first-attempt success rate, complaint flag), we will use two-proportion z-tests. These analyses are exploratory and will be identified as secondary checks.

3.5 Regression Estimation

To improve precision and adjust residual imbalance, we estimate the following intent-to-treat regression models:

Continuous outcomes (e.g., Route Duration):

$$Y_{it} = \alpha + \beta \cdot \text{Treatment}_{it} + \delta X_{it} + \gamma_t + \epsilon_{it}$$

where Y_{it} is the outcome for driver i on shift t; Treatment_{it} is treatment assignment; X_{it} are time-varying controls (parcel volume, distance, weather); γ_t are week fixed effects capturing temporal shocks. Standard errors clustered at the driver level account for repeated measures across shifts.

Binary outcomes (e.g., OnTime, Complaint):

$$P(Y_{it} = 1) = \text{logit}^{-1}\left(\alpha + \beta \cdot \text{Treatment}_{it} + \delta X_{it} + \gamma_t\right)$$

Average marginal effects will be reported (percentage-point changes attributable to treatment) for managerial interpretability.

We will present both parsimonious models (treatment only) and full adjusted models (treatment + controls) to show robustness of β.

3.6 Manipulation Check

For each shift, we compute **Reorder Flag = 1** if actual stop sequence $\neq$ planned sequence within next-three window. We compare group means (t-test or proportion test). We expect a substantially higher rate in treatment; failure to observe this would prompt interpretation caution and a focus on per-protocol results.

3.7 Outliers, Missing Data, and Data Quality

- **Extreme Durations:** Shifts beyond ±3 SD of group mean will be flagged; main ITT includes them; a robustness run excludes flagged shifts.
- **Partial Shifts / Device Failure:** If GPS or timestamps are missing for >20% of stops in a shift, that shift is excluded from primary analysis.
- **Survey Nonresponse:** Customer Rating is missing when no response; analyses using ratings will be descriptive. Service inference relies primarily on On-Time % and Complaint Flag, which are observed operationally.

3.8 Heterogeneity Exploration

We will explore whether treatment effects vary across prespecified subgroups: driver tenure (¡1 year vs. $\geq$ 1 year), route environment, and stop density. Estimate interaction models (Treatment $\times$ Subgroup) in OLS/logit; display marginal effects with CIs. These analyses are hypothesis-generating; we will not draw firm conclusions without followup tests.

3.9 Translating Effects to Business Impact

We will convert statistically estimated time savings into annualized labor and fuel savings under plausible scaling assumptions. Example calculation: if Route Duration falls by Δ minutes per shift, and the company operates D driver-shifts per year in comparable regions, annual hours saved = $(\Delta/60) \times D$. Multiplying by blended labor + vehicle cost per hour yields a monetary benefit estimate. Comparing that to one-time app enhancement and training costs yields an ROI ratio. We will also estimate incremental delivery capacity (additional stops/day) that could support revenue growth or service improvements without extra headcount.

3.10 Decision Rules

We developed three decision rules to determine the actions to take after experiment results are in:

- **Go / Scale:** Statistically significant Route Duration reduction and no meaningful decline (>1–2 pp) in On-Time Delivery %.
- **Pilot Extension / Targeted Rollout:** Small but positive efficiency gain with heterogeneous effects—deploy in route types that benefit most.

- **No Scale:** Efficiency gain negligible (<1 min) or service degradation > preset guardrail.

These explicit rules help management act quickly once results are in.

4 Challenges and Mitigation

- **Driver compliance:** Some treatment drivers may underuse/overuse the reordering autonomy, or some control drivers may ignore instructions. We will mitigate this by clear training and communication. GPS logs will monitor compliance: random audits will verify that treatment drivers engage with reordering and that control drivers largely follow their plan. Emphasizing that the trial is for learning (not a performance review) should encourage honest participation.

- **Traffic and route variability:** Different routes and congestion may introduce noise. Stratification ensures balance on known factors (urban/suburban, stop density). Regression models will control for route characteristics and day-of-week effects. The eight-week period captures varied conditions (rush hour, midday, weekday, weekend). Outlier days (e.g., strikes, disasters) will be flagged and excluded or modeled separately in robustness checks.

- **Measuring customer satisfaction:** Survey response rates might be low and ratings are subjective. To address this, we will supplement surveys with objective proxies: on-time delivery percentage and complaint incidence. We assume survey nonresponse is random across groups due to random assignment. We will report response rates and, if necessary, restrict some analyses to routes with valid feedback (while noting reduced power).

- **External validity:** Conducting the study in one urban depot limits generalizability, which we will acknowledge. Internal validity is supported by randomization and stratification. We will compare sample demographics (e.g., age, tenure) to the company-wide population to assess representativeness.

- **Time trends and attrition:** Driver behavior may change over time (learning effects), and some drivers might drop out. We will check whether attrition (if any) is balanced across groups; if a driver exits, their data will be included up to that point (intention-to-treat). We also expect some initial adjustment; a brief run-in period (e.g., excluding the first few shifts) may be used as a sensitivity check.

- **Statistical assumptions:** We assume randomization created comparable groups. We will test balance on key covariates at baseline. If

a severe imbalance appears, we will adjust by covariate regression or matching. We will ensure all statistical tests meet their assumptions or else use robust/nonparametric methods.

By planning these checks and mitigations, we aim to isolate the causal impact of granting drivers limited autonomy to reorder stops. Any group differences can then be confidently attributed to this flexibility's effect on operational efficiency and customer service.